

Algorithmic Foundations and Protocol Invariants for the Avalanche Matchmaking Protocol (AMP v2)

Quorum Intersection Proofs, State-Exhaustive Value Conservation, and Game-Theoretic Settlement Analysis

DRAFT — internal spec snapshot. Not for external distribution.

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Abstract

This paper establishes the mathematical specification and algorithmic invariants for the Avalanche Matchmaking Protocol (AMP v2). We analyze the safety, economic finality, and queue dynamics of decentralized, oracle-free N -player match settlement. Specifically, we prove: (1) Byzantine fault-tolerant K -of- N quorum intersection under non-equivocation assumptions across all residue classes ($N \geq 3f + 1$); (2) state-exhaustive integer conservation of pooled stakes down to 1 wei across all terminal match states (Settled, TimeoutClaim, FundingExpired, SilenceRefund, DisputeResolved); (3) permutation invariance of multi-opponent Glicko-2 field updates over $\mathbb{R}$ with a canonically ordered computational model; (4) game-theoretic dominance of dual-deposit reporting bonds in pivotal dropout scenarios; and (5) conditional uniformity bounds of candidate lobby assignment.

Keywords: Byzantine Quorum Systems, Mechanism Design, Decentralized Matchmaking, Glicko-2, Value Conservation, Avalanche L1 Subnets.

1. Byzantine Quorum Intersection

Let an N -player match session be represented by participants $\mathcal{P} = \mathcal{H} \cup \mathcal{B}$, where $\mathcal{H}$ denotes honest participants, $\mathcal{B}$ denotes Byzantine or colluding participants, and $|\mathcal{H} \cap \mathcal{B}| = 0$. We bound the Byzantine population by $|\mathcal{B}| \leq f$.

Definition 1 (Quorum Threshold).

For a lobby of size N , canonical ladder settlement requires a concordant quorum of K participant signatures:

$$K = \lfloor 2N/3 \rfloor + 1, \quad K = N \text{ for } N \leq 3$$

Each participant holds an ECDSA keypair (sk_i, pk_i) . An attestation commits to the EIP-712 session digest over the structured message

$$m = \text{keccak256}(\text{matchId} \parallel \text{chainId} \parallel \text{sessionNonce} \parallel \vec{R} \parallel H_{\text{transcript}})$$

where $\vec{R}$ is the ranked placement vector and $H_{\text{transcript}}$ is the deterministic execution trace hash.

Theorem 1 (Quorum Non-Equivocation).

For any $N \geq 3f + 1$, two conflicting match states $m \neq m'$ cannot both achieve a valid quorum of size K without at least one Byzantine participant producing two distinct signatures.

Proof. Let $Q_1, Q_2 \subseteq \mathcal{P}$ be two valid quorums supporting conflicting states $m \neq m'$, with $|Q_1| \geq K$ and $|Q_2| \geq K$. By inclusion-exclusion:

$$|Q_1 \cap Q_2| = |Q_1| + |Q_2| - |Q_1 \cup Q_2| \geq 2K - N$$

We evaluate $2K - N$ across all residue classes $N = 3f + r$ where $r \in \{1, 2, 3\}$:

- **Case 1** ($r = 1, N = 3f + 1$): $K = \lfloor (6f + 2)/3 \rfloor + 1 = 2f + 1$, hence

$$|Q_1 \cap Q_2| \geq 2(2f + 1) - (3f + 1) = f + 1.$$

- **Case 2** ($r = 2, N = 3f + 2$): $K = \lfloor (6f + 4)/3 \rfloor + 1 = 2f + 2$, hence

$$|Q_1 \cap Q_2| \geq 2(2f + 2) - (3f + 2) = f + 2 \geq f + 1.$$

- **Case 3** ($r = 3, N = 3f + 3$): $K = \lfloor (6f + 6)/3 \rfloor + 1 = 2f + 3$, hence

$$|Q_1 \cap Q_2| \geq 2(2f + 3) - (3f + 3) = f + 3 \geq f + 1.$$

Thus, $|Q_1 \cap Q_2| \geq f + 1$ holds for all $N \geq 3f + 1$. Because the total Byzantine population is bounded by $|\mathcal{B}| \leq f$, the intersection contains at least one honest participant:

$$|(Q_1 \cap Q_2) \cap \mathcal{H}| \geq (f + 1) - f = 1$$

Let $p^* \in (Q_1 \cap Q_2) \cap \mathcal{H}$. By definition of an honest node, p^* signs at most one digest per session nonce:

$$\sigma_{p^*}(m) \Rightarrow \neg \exists \sigma_{p^*}(m') \quad \text{for } m' \neq m$$

Therefore, two conflicting quorums cannot both exist unless at least one Byzantine node $p \in \mathcal{B}$ equivocally signs both m and m' . Submitting both signatures on-chain constitutes non-repudiable proof of fraud, slashing p . ■

Remark (Liveness Bound & Threat Boundary).

At $N = 3f + 1$, honest participant count $|\mathcal{H}| = 2f + 1 = K$. A single honest crash-fault ($|\mathcal{H}| - 1$) starves immediate concordant quorum, triggering the optimistic grace timeout.

Furthermore, Theorem 1 guarantees **consensus non-equivocation**, not game correctness: if K Byzantine players collude to sign a fraudulent ladder with matching $H_{\text{transcript}}$, the contract accepts it unless an honest minority escalates inputs to the optimistic dispute challenge window.

2. State-Exhaustive Value Conservation

Match escrow must satisfy exact mathematical conservation across all terminal execution states to prevent protocol insolvency.

Definition 2 (Dual-Deposit Escrow and Per-State Inflow).

Every participant $i \in \mathcal{P}$ deposits stake S and reporting bond B_{rep} , so $D_i = S + B_{\text{rep}}$ and the deposit base is

$$\mathcal{E}_{\text{deposits}} = \sum_{i=1}^N D_i = NS + NB_{\text{rep}}.$$

If a dispute is opened, each challenging faction additionally escrows a challenge stake C_c at challenge time; the **inflow to a terminal state** s is the value actually escrowed entering that state:

$$\mathcal{E}_{\text{in}}(s) = \begin{cases} NS + NB_{\text{rep}} + \sum_c C_c & \text{if a dispute was opened} \\ NS + NB_{\text{rep}} & \text{otherwise.} \end{cases}$$

Conservation claims are always relative to $\mathcal{E}_{\text{in}}(s)$ — never to a fixed constant.

Let $\Omega_{\text{gross}} = NS$. Gross protocol fee at basis points $\beta_{\text{rake}} \in [0, 10000]$ is:

$$F_{\text{gross}} = \left\lfloor \frac{\Omega_{\text{gross}} \beta_{\text{rake}}}{10000} \right\rfloor, \quad \Omega_{\text{net}} = \Omega_{\text{gross}} - F_{\text{gross}}$$

$$F_{\text{studio}} = \left\lfloor \frac{F_{\text{gross}} \beta_{\text{studio}}}{10000} \right\rfloor, \quad F_{\text{protocol}} = F_{\text{gross}} - F_{\text{studio}}$$

Protocol Invariant 1 (Zero-Loss Division Remainder).

For T prize tiers ($\{t_1, \dots, t_T\}$, $\sum t_j = 10000$), preliminary payouts $P_j = \left\lfloor \frac{\Omega_{\text{net}} \cdot t_j}{10000} \right\rfloor$ leave remainder $\rho = \Omega_{\text{net}} - \sum_{j=1}^T P_j$. By construction:

$$0 \leq \rho < T \text{ wei}$$

The implementation routes ρ to the protocol reserve as a fee, so

$$\sum_{j=1}^T P_j + \rho = \Omega_{\text{net}}$$

holds exactly and no value is stranded.

Theorem 2 (State-Exhaustive Conservation).

In every terminal match state $s \in \mathcal{T}_{\text{states}} = \{\text{Settled}, \text{TimeoutClaim}, \text{FundingExpired}, \text{SilenceRefund}, \text{DisputeResolved}\}$, total ledger credits equal the per-state inflow:

$$\sum \text{Outflows}(s) = \mathcal{E}_{\text{in}}(s) \quad \text{to 1 wei.}$$

Proof. We evaluate outflows across all five terminal states. Payout delivery is lazy (prove-your-payout claims); the identities hold for the **ledger** the moment the terminal state is recorded, since every credit below is either written at resolution or is a deterministic function of the recorded ladder and immutable tier profile.

- **State 1 (Settled):** Quorum $|\mathcal{S}| \geq K$ reached during the quorum window. Net pool distributes via normalized tiers (plus remainder $\rho = \mathcal{P}$ to the reserve); signers receive B_{rep} back; non-signers $\mathcal{S}$ have their bonds slashed and split 50/50 between the relayer fund and Rank 1:

$$\begin{aligned} \sum \text{Out} &= (\Omega_{\text{net}} - \rho) + \rho + F_{\text{gross}} + |\mathcal{S}| B_{\text{rep}} \\ &\quad + \left\lfloor \frac{|\mathcal{M}| B_{\text{rep}}}{2} \right\rfloor + \left(|\mathcal{M}| B_{\text{rep}} - \left\lfloor \frac{|\mathcal{M}| B_{\text{rep}}}{2} \right\rfloor \right) \\ &= NS + (|\mathcal{S}| + |\mathcal{M}|) B_{\text{rep}} = NS + NB_{\text{rep}} = \mathcal{E}_{\text{in}}(s). \end{aligned}$$

- **State 2 (TimeoutClaim):** The quorum window lapses; the ladder's Rank-1 participant posts a unilateral claim, unchallenged through the grace window. **Slash rule (as shipped):** every participant *except the claimant* forfeits their bond – the claimant's signature is the only one recorded in this state, so “all bonds but the claimant's” is exactly the non-signer set of the recorded settlement. The claimant recovers their own bond plus the winners' half of the $N - 1$ slashed bonds (the relayer takes the other half):

$$\begin{aligned} \sum \text{Out} &= \Omega_{\text{net}} + F_{\text{gross}} + B_{\text{rep}} + (N - 1) B_{\text{rep}} \\ &= NS + NB_{\text{rep}} = \mathcal{E}_{\text{in}}(s). \end{aligned}$$

- **State 3 (FundingExpired):** The lobby fails to fill before the escrow deadline; only the set J of joined participants ($|J| < N$) ever deposited, so the per-state inflow is $\mathcal{E}_{\text{in}}(s) = |J| (S + B_{\text{rep}})$. Each joined participant claims a full refund:

$$\sum \text{Out} = \sum_{j \in J} (S + B_{\text{rep}}) = |J| (S + B_{\text{rep}}) = \mathcal{E}_{\text{in}}(s).$$

- **State 4 (SilenceRefund):** The lobby filled but the quorum and grace windows lapse with **no** claim at all (mass disconnect). No rake, no slash; every deposit refunds:

$$\sum \text{Out} = \sum_{i \in \mathcal{P}} (S + B_{\text{rep}}) = NS + NB_{\text{rep}} = \mathcal{E}_{\text{in}}(s).$$

- **State 5 (DisputeResolved):** A ≥ 2 -signer concordant challenge contested the grace claim; the bonded verifier ruled, and the challenge stakes $\sum_c C_c$ are part of the inflow per Definition 2. Disputes take no rake; the valid ladder's tiers are paid over the gross stake pool NS (losers' stakes are thereby consumed by the ladder itself); non-losers recover bonds; the losers' bonds plus all challenge stakes form the slash pool, split 70% to the winning faction and 30% to the protocol reserve (rounding dust to the reserve):

$$\begin{aligned} \sum \text{Out} &= \underbrace{NS}_{\text{tiers over gross}} + \underbrace{(N - \ell) B_{\text{rep}}}_{\text{non-loser bonds}} \\ &\quad + \underbrace{\left(\ell B_{\text{rep}} + \sum_c C_c \right)}_{\text{slash pool, fully split}} \\ &= NS + NB_{\text{rep}} + \sum_c C_c = \mathcal{E}_{\text{in}}(s), \end{aligned}$$

where ℓ is the losing faction's size.

Conservation holds to 1 wei across all terminal states, relative to the value actually escrowed in each. ■

3. Multi-Opponent Rating Invariance

To update skill ratings in fields of N players, `amp-match-core` executes `glicko2_update_vs_many` across all $N - 1$ opponents within a single rating period.

Definition 3 (Field Outcomes & Glicko-2 Formulation).

For player i facing opponent $j \neq i$, outcome score s_{ij} is:

$$s_{ij} = \begin{cases} 1.0 & \text{if } \text{rank}(i) < \text{rank}(j) \\ 0.5 & \text{if } \text{rank}(i) = \text{rank}(j) \\ 0.0 & \text{if } \text{rank}(i) > \text{rank}(j) \end{cases}$$

With scale transformations $\mu = (r - 1500)/173.7178$ and $\varphi = \text{RD} / 173.7178$ (where RD is the rating deviation), let $E_{ij} = (1 + \exp(-g(\varphi_j)(\mu_i - \mu_j)))^{-1}$ and $g(\varphi) = (1 + 3\varphi^2/\pi^2)^{-1/2}$:

$$v_i = \left(\sum_{j \neq i} g(\varphi_j)^2 E_{ij} (1 - E_{ij}) \right)^{-1}, \quad \Delta_i = v_i \sum_{j \neq i} g(\varphi_j) (s_{ij} - E_{ij})$$

Theorem 3 (Permutation Invariance over $\mathbb{R}$).

Let $\pi : \{1, \dots, N - 1\} \rightarrow \{1, \dots, N - 1\}$ be any permutation of opponent processing order. The mathematical tuple $(\mu_{i'}, \varphi_{i'}, \sigma_{i'}) \in \mathbb{R}^3$ is invariant under π .

Proof. Functions $\psi(i, j) = g(\varphi_j)^2 E_{ij} (1 - E_{ij})$ and $\omega(i, j) = g(\varphi_j) (s_{ij} - E_{ij})$ depend strictly on the unordered pair $\{i, j\}$. Over the reals, addition is commutative and associative:

$$\sum_{k=1}^{N-1} \psi(i, \pi(k)) = \sum_{j \neq i} \psi(i, j), \quad \sum_{k=1}^{N-1} \omega(i, \pi(k)) = \sum_{j \neq i} \omega(i, j)$$

Thus, scalar field parameters v_i and Δ_i are invariant under π . Volatility $\sigma_{i'}$ is the unique zero of the objective function:

$$f(x) = \frac{e^x (\Delta_i^2 - \varphi_i^2 - v_i - e^x)}{2(\varphi_i^2 + v_i + e^x)^2} - \frac{x - \ln \sigma_i^2}{\tau^2}$$

Because $f(x)$ is parameterized entirely by invariant scalars, its root x^* is invariant under π . Derived quantities $\varphi_{i'}$ and $\mu_{i'} = \mu_i + (\varphi_{i'})^2 \sum \omega(i, j)$ are identical over $\mathbb{R}$. ■

Remark (Computational Model & Modeling Assumptions).

Floating-point addition (IEEE-754) is non-associative. To guarantee bit-identical updates across different node architectures, `amp-match-core` enforces a **canonical accumulation order**: opponent contributions are sorted by the opponent tuple's IEEE-754 bit patterns (rating, then deviation, then score) before summation, making the floating-point sum a deterministic function of the opponent **multiset**. This property is fuzz-gated: any permutation of an opponent field must yield bit-identical $(\mu', \varphi', \sigma')$.

Modeling Note: Treating an N -player ranking as $N - 1$ independent pairwise matches is a standard Glickman rating-period approximation. Because tournament ranks are dependent, this models directional skill delta accurately but contracts φ faster than a joint multi-variate model.

4. Game-Theoretic Settlement Analysis

We evaluate the strategic incentives of an eliminated player k who placed outside the payout tiers ($r_k > T$) and considers withholding their signature to cause a match timeout.

Definition 4 (Pivotal Dropout Game Parameters).

Let $S > 0$ be the match stake, $B_{\text{rep}} > 0$ be the reporting bond, and $C_{\text{sign}} \approx 0$ be client signing gas. A player is **pivotal** if their refusal to sign reduces total signatures below quorum ($|\mathcal{S}| = K - 1$).

Theorem 4 (Dominance of Cooperation under Timeout Slashing).

Under the protocol's **TimeoutSlash** rule, signing the exit attestation strictly dominates defecting whenever player k 's cooperation is pivotal or quorum would otherwise be reached, and weakly dominates in the residual case where quorum fails regardless of k 's action.

Proof. Under protocol rules, a match timeout does **not** refund stake S to uncooperative participants. If quorum fails:

1. The match enters the `TimeoutClaim` state post-grace.
2. The highest-attested valid survivor (Rank 1) claims net escrow Ω_{net} .
3. Per the State-2 slash rule, **every participant except the claimant** forfeits B_{rep} — including any player who signed but whose signature did not enter a recorded settlement. (Equivalently: bond recovery requires being the recorded signer of a **settling** ladder; a signature that never settles recovers nothing.)

We evaluate player $k \neq$ claimant's terminal payoff matrix:

- **Scenario A (quorum reached without k , $|\mathcal{S} \setminus \{k\}| \geq K$):**

$$U_k(\text{Sign}) - U_k(\text{Defect}) = (-S + B_{\text{rep}} - C_{\text{sign}}) - (-S - B_{\text{rep}}) = 2B_{\text{rep}} - C_{\text{sign}} > 0.$$

- **Scenario B (k strictly pivotal, $|\mathcal{S} \setminus \{k\}| = K - 1$):** If k cooperates the ladder settles with k in the signer set:

$$U_k(\text{Sign}) = -S + B_{\text{rep}} - C_{\text{sign}}.$$

If k defects, quorum starves, the match settles via `TimeoutClaim`, and k (a non-claimant) is slashed:

$$U_k(\text{Defect}) = -S - B_{\text{rep}}. \quad U_k(\text{Sign}) - U_k(\text{Defect}) = 2B_{\text{rep}} - C_{\text{sign}} > 0.$$

- **Scenario C (quorum fails regardless of k):** `TimeoutClaim` slashes every non-claimant whether or not they signed, so $U_k(\text{Sign}) = U_k(\text{Defect}) = -S - B_{\text{rep}}$ and signing is weakly dominant (the $C_{\text{sign}} \approx 0$ gap is one-directional only through the claimant's tip, which k forgoes either way).

Because defection never triggers a stake refund and always forfeits B_{rep} whenever it matters, cooperation dominates defection across all game states. ■

Remark (Scope of Theorem 4).

Theorem 4 is a corollary of the State-2 slash rule, not the protocol's economic centerpiece: it states that a reporting bond of any positive size makes exit-signing incentive-compatible for losing, non-pivotal players. It does not model collusion among $\geq K$ signers (covered by Theorem 1's fraud evidence) or verifier capture in disputes (covered by the bonded-verifier stake).

5. Queue Allocation & Entropy Bounds

To prevent pre-coordinated collusion in open staked FFA queues, AMP partitions the matchmaking queue using a commit-reveal shuffle.

Definition 5 (Commit-Reveal Shuffle).

Candidates submit blinded commitments $H_i = \text{keccak256}(p_i \parallel S \parallel \text{salt}_i)$. At candidate pool size $M \geq N$, the coordinator commits to target block height $H_{\text{target}} = H_{\text{current}} + \Delta$ ($\Delta \geq 2$). After salts are revealed, the assignment seed is

$$\Xi = \text{keccak256}(\mathcal{B}_{\text{target}} \parallel \text{salt}_1 \parallel \text{salt}_2 \parallel \dots \parallel \text{salt}_R)$$

over the R revealed salts, where $\mathcal{B}_{\text{target}}$ is the Avalanche blockhash at H_{target} .

Theorem 5 (Conditional Allocation Uniformity).

Conditioned on a seed Ξ drawn uniformly, $\Xi \sim \text{Uniform}(\{0, 1\}^{256})$, a deterministic Fisher-Yates shuffle achieves uniform distribution over candidate permutations. For any two candidates $i \neq j$, lobby co-location probability is:

$$\mathbb{P}(\text{Lobby}(i) = \text{Lobby}(j)) = \frac{\binom{M-2}{N-2}}{\binom{M-1}{N-1}} = \frac{N-1}{M-1}$$

Proof. Under a uniform random permutation of M candidates into a lobby of size N , the number of subsets of size N containing both i and j is $\binom{M-2}{N-2}$. Total possible size- N lobbies from pool M is $\binom{M-1}{N-1}$ (fixing candidate i , choose the remaining $N-1$ from $M-1$). The ratio yields $\frac{N-1}{M-1}$.

For a colluding cartel of size c , co-locating all c members in a single lobby decays as:

$$\mathbb{P}(\text{Cartel Co-Location}) = \prod_{k=1}^{c-1} \frac{N-k}{M-k} \in \mathcal{O}\left(\left(\frac{N}{M}\right)^{c-1}\right)$$

Remark (Proposer Discretion & Threat Boundaries).

Avalanche Snow consensus provides sub-second finality, but single-block proposers retain transaction-ordering and block-withholding discretion. True unbiasedness of $\mathcal{B}_{\text{target}}$ is therefore an engineering approximation: Theorem 5 is **conditional** on the seed's uniformity, which is assumed, not proven.

Mitigations: (1) Reveal deadlines slash deposits of unrevealed commitments, preventing selective aborts; (2) The production roadmap transitions Ξ to threshold BLS beacons via Avalanche Warp Messaging (AWM).

6. Machine-Checked Invariants & Fuzz Gates

Protocol invariants are continuously asserted via stateful property testing and fuzzing:

Protocol Invariant	Formal Bound	Automated Test Gate
Quorum Non-Equiv.	$ Q_1 \cap Q_2 \geq f + 1$	Conflicting-quorum terminal test
State Conservation	$\sum \text{Out} = \mathcal{E}_{\text{in}}(s)$ to 1 wei	Foundry invariant fuzz (256 runs x 128k calls)
Glicko π -Invariance	Bit-identical $(\mu', \varphi', \sigma')$	proptest permutation suite (10k cases)
Pull Accounting	Zero external calls in resolution	Reverting-receiver isolation test
Shuffle Uniformity	Lemire-debiased bounded draws	Distribution spread test (7k draws/bucket)

These gates **test** the invariants; they are not proofs of Theorems 2 or 4. The conservation fuzz enforces the Theorem-2 identities mechanically across randomized op sequences; the quorum and payoff identities are proven above.

7. Gas Execution Bounds

On-chain verification in `AMPMultiplayer.sol` records the settlement (constant storage) and verifies K packed signatures; per-player payouts are claimed lazily, so settlement gas scales with signatures, not recipients:

Lobby (N)	Quorum (K)	Measured settlement gas
8 players	6 sigs	152,962 gas (spec gate: under 250k)
16 players	11 sigs	195,844 gas
64 (BR)	43 sigs	404,021 gas

By restricting balance distribution to internal ledger credit (`claimable[player] += reward`), 64-player Battle Royale settlements avoid iterative external call overhead, remaining well within Avalanche block gas limits (30,000,000 gas).

8. Deployments and Verification Status

All contract deployments live on the Avalanche **Fuji testnet**. Nothing is deployed to mainnet.

- **v1 1v1 settlement stack (live on Fuji, Sourcify exact-match verified):** AMPRegistry at `0xf6B0eA6c88c574c4BbEAdC186AAfe72C43C2cDc2` and AMPSettlement at `0x78ec93e66255a74873d20DD62C6595A389272126`.
- **v2 N-player engine (live on Fuji, Sourcify full-match verified):** AMPMultiplayer at `0xcabf7b626172fE55d54f03c346563671AbcC77f7`, branch `n-player-multiplayer` (5bbc816). Implements dual-deposit escrow, bitmask quorum verification, and immutable payout profiles with lazy claims.
- **Match core library (amp-match-core):** Pure Rust core implementing `party.rs`, team/FFA/BR queue topologies, commitments and the blockhash shuffle, and canonical Glicko-2 field updates.

Engineering Sign-Off (draft):

The invariants specified here are implemented in `AMPMultiplayer.sol` and `amp-match-core` and exercised by 91 Foundry tests (including the two stateful conservation invariants) and 106 Rust tests (including the permutation and party-graph fuzz gates). Fuzz results are empirical evidence, not proof; the proofs in this document stand on the stated assumptions. External audit and timelock governance precede any mainnet deployment.